MVP DEVELOPER PACKAGE

Evolve Edge
Developer Deliverables

Complete specifications for Permission Matrix, Tier Comparison, Development Schedule & API Endpoints — ready for developer handoff.

VERSION	DATE	STATUS
1.0	March 2026	Ready for Handoff

STACK

Next.js · FastAPI · AWS

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DELIVERABLE 01

Role-Based Permission Matrix

Defines exactly what Admin, Coach, and Athlete roles can view, create, edit, and delete across every application module.

Legend: ✓ = Full access ✗ = No access Cond. = Conditional / scoped access (see note) N/A = Not applicable to this role

Authentication & Account Management

Action	Admin	Coach	Athlete
Register account	✓	✓	✓
Login / Logout	✓	✓	✓
Reset own password	✓	✓	✓
Enable / disable MFA (own)	✓	✓	✓
View & edit own profile	✓	✓	✓
Close own account	✓	✓	✓
Reopen / restore any account	✓	✗	✗
Force logout another user	✓	✗	✗

User Management

Action	Admin	Coach	Athlete
View all users (platform-wide)	✓	✗	✗
Invite new Coach	✓	✗	✗
Invite new Athlete	✓	◂ Own roster	✗
Edit any user profile	✓	✗	✗
Assign / change user roles	✓	✗	✗
Deactivate / archive user	✓	✗	✗
Delete user account	✓	✗	✗
View & manage own roster (athletes)	✓	✓	✗

Athlete Profiles

Action	Admin	Coach	Athlete
View all athlete profiles	✓	✗	✗
View assigned athletes' profiles	✓	✓	✗
View own profile	✓	✓	✓
Edit assigned athlete's profile	✓	◂ Assigned only	✗
Add / update biometric data	✓	◂ Assigned only	◂ Own only
Set athlete goals	✓	◂ Assigned only	◂ Own only
Export athlete profile (PDF)	✓	◂ Assigned only	◂ Own only

Workout Library & Exercise Database

Action	Admin	Coach	Athlete
View global exercise library	✓	✓	✓
Create new exercise	✓	✓	✗
Edit any exercise	✓	✗	✗
Edit own created exercise	✓	✓	✗
Upload exercise media (video/image)	✓	✓	✗
Create workout template	✓	✓	✗
Edit / delete own template	✓	✓	✗
Edit any workout template	✓	✗	✗
View assigned templates	✓	✓	◂ Assigned only
Duplicate / clone a template	✓	✓	✗

Workout Assignment & Scheduling

Action	Admin	Coach	Athlete
Assign workout to athlete	✓	◂ Own roster	✗
Schedule workout (date/time)	✓	✓	✗
View own assigned schedule	✓	✓	✓
View athlete's assigned schedule	✓	◂ Own roster	✗
Modify / cancel assigned workout	✓	◂ Own roster	✗
Mark rest / recovery day	✓	✓	◂ Own only

Workout Logging

Action	Admin	Coach	Athlete
Log own workout session	✓	✓	✓
Log workout on behalf of athlete	✓	◂ Own roster	✗
Edit own log entry	✓	✓	✓
Edit athlete's log entry	✓	◂ Own roster	✗
Delete own log entry	✓	✓	✓
Delete any athlete's log entry	✓	✗	✗
View own workout history	✓	✓	✓
View athlete's workout history	✓	◂ Own roster	✗
Add session notes / RPE	✓	✓	✓
Attach media to log entry	✓	✓	✓

Progress Tracking & Analytics

Action	Admin	Coach	Athlete
View platform-wide analytics	✓	✗	✗
View own progress charts	✓	✓	✓
View assigned athlete's progress	✓	◂ Own roster	✗
Generate & export PDF reports	✓	◂ Own roster	◂ Own only
Set / modify progress goals	✓	◂ Own roster	◂ Own only
View athlete comparison / leaderboard	✓	✓	✗

Notifications

Action	Admin	Coach	Athlete
Receive transactional & security alerts	✓	✓	✓
Send platform-wide announcement	✓	✗	✗
Send message to athlete	✓	◐ Own roster	✗
Send message to coach	✓	✗	✓
Manage own notification preferences	✓	✓	✓

Billing & Subscriptions

Action	Admin	Coach	Athlete
View own subscription plan	✓	✓	✓
Upgrade / downgrade plan	✓	✓	✓
Manage payment method & invoices	✓	✓	✓
View all org billing (platform-wide)	✓	✗	✗
Issue refunds	✓	✗	✗

Admin Portal & System Settings

Action	Admin	Coach	Athlete
Access Admin Portal	✓	✗	✗
View audit logs (2-year retention)	✓	✗	✗
Configure global settings	✓	✗	✗
Manage exercise / framework library	✓	✗	✗
View platform usage metrics	✓	✗	✗
Configure subscription tier limits	✓	✗	✗

DELIVERABLE 02

Free vs Premium Tier Comparison

Defines feature gates per subscription plan — required for Stripe configuration and developer feature-flag implementation.

Developer Note: All features marked PREMIUM must be gated behind a subscription check middleware. Free tier limits marked LIMITED must return HTTP 403 with `"error": "upgrade_required"` when exceeded.

Plan Overview

Category	Free	Premium
Monthly Price	$0 / month	$X / month (TBD)
Target User	Solo athletes, trial users	Coaches, serious athletes, teams
Support	Community / Email	Priority email + in-app chat

User & Roster Limits

Feature	Free	Premium
Athletes per coach	3 max	✓ Unlimited
Organization / team accounts	✗	✓
Multi-coach organization	✗	✓ Add-on
External collaborator invites	✗	✓

Athlete Profile

Feature	Free	Premium
Basic profile (name, age, bio, avatar)	✓	✓
Biometric & health data tracking	3 metrics max	✓ Unlimited
Active goals	1 goal max	✓ Unlimited
Custom profile fields	✗	✓
Profile export (PDF)	✗	✓

Workout Library & Templates

Feature	Free	Premium
Access global exercise library	✓	✓
Create custom exercises	5 max	✓ Unlimited
Create workout templates	3 max	✓ Unlimited
Upload exercise media (video/image)	✗	✓
Duplicate / clone templates	✗	✓
Template tagging & categorisation	✗	✓
Share templates with athletes	1 template	✓ Unlimited

Workout Assignment & Scheduling

Feature	Free	Premium
Assign workouts to athletes	Max 3 athletes	✓ Unlimited
Calendar scheduling view	Week view only	✓ Full calendar
Recurring workout scheduling	✗	✓
Training block / periodisation	✗	✓
Rest day & recovery scheduling	✓	✓

Workout Logging

Feature	Free	Premium
Log own workouts (sets/reps/weight)	✓	✓
Log RPE & session notes	✓	✓
Attach media to log entry	✗	✓
Coach log on behalf of athlete	✗	✓
Workout log history retention	30 days only	✓ Unlimited
Edit past log entries	Within 24 hrs	✓ Unlimited

Progress Tracking & Analytics

Feature	Free	Premium
View own progress charts (basic)	✓ Basic	✓ Advanced
Strength trend visualisation	1 metric only	✓ All metrics
Volume load tracking	✗	✓
Body composition tracking	✗	✓
Athlete comparison / leaderboard	✗	✓
Historical trend range	Last 7 days	✓ All time
Generate & export PDF reports	✗	✓
Automated weekly summary email	✗	✓

Notifications

Feature	Free	Premium
In-app & email (transactional)	✓	✓
Workout reminders (email)	✗	✓
Weekly coach summary emails	✗	✓
Custom notification preferences	✗	✓
SMS & Slack / Teams (future)	✗	Roadmap

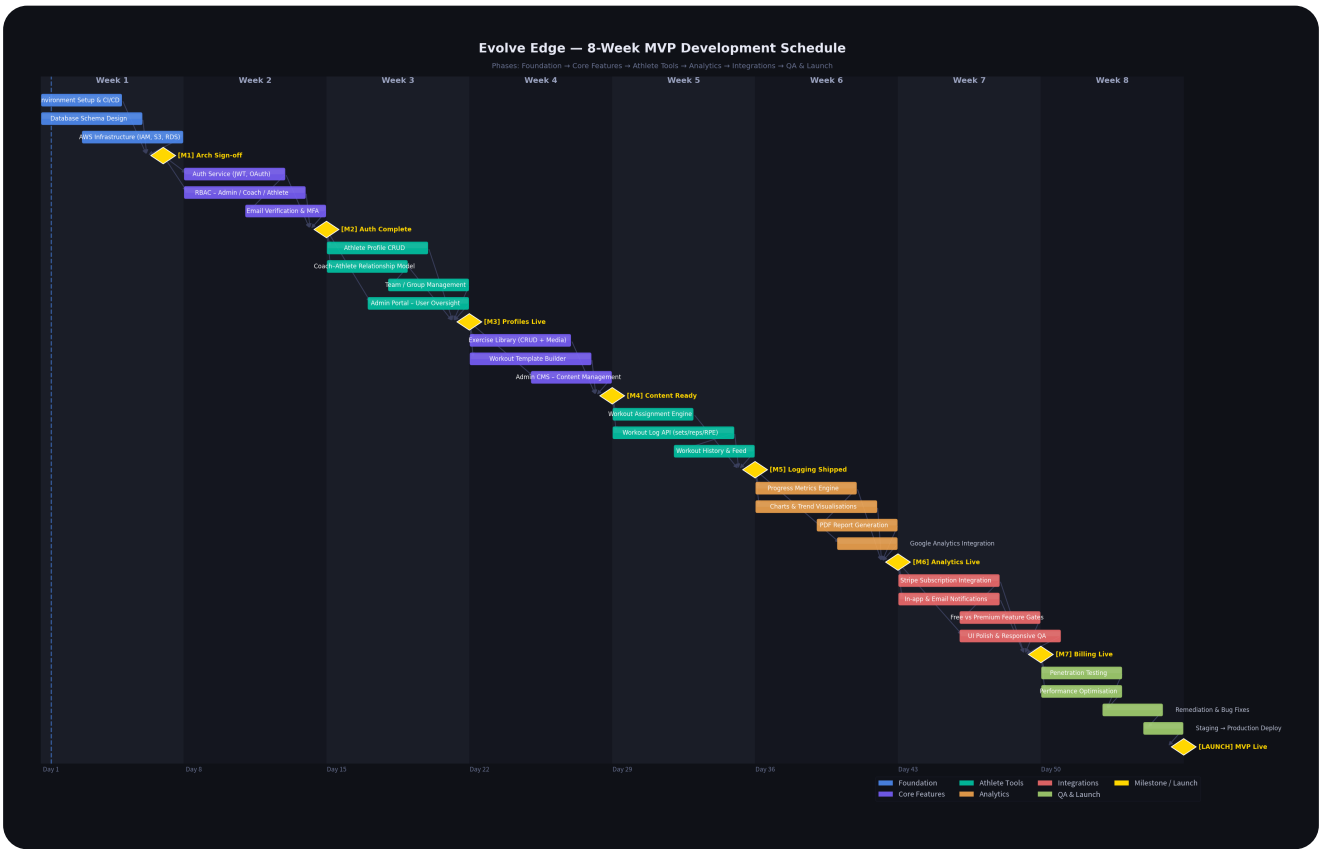
Security & Data

Feature	Free	Premium
AES-256 encryption & TLS 1.2+	✓	✓
Social login (Google / Microsoft)	✓	✓
Optional MFA	✓	✓
Audit log access	✗	✓
Data export / portability	✗	✓
Data retention period	30 days post-closure	✓ 1 year

DELIVERABLE 03

8-Week Development Schedule

Visual Gantt chart with phased milestones, task dependencies, and weekly gates. Each milestone is a hard blocker for the following phase.



Milestone Gate Summary

M1 **Architecture Sign-off**
End of Week 1

AWS provisioned · DB schema approved · CI/CD pipeline green · All 3 foundation tasks passing.

M2 **Auth Complete**
End of Week 2

JWT auth working · RBAC enforced for all 3 roles · MFA via OTP & Authenticator · Social login (Google, Microsoft) passing.

M3 **Profiles Live**
End of Week 3

Athlete CRUD complete · Coach–Athlete relationships functional · Team management operational · Admin user oversight live.

M4 **Content Ready**
End of Week 4

Exercise library with media upload · Workout template builder · CMS for dynamic assets operational.

M5 **Logging Shipped**
End of Week 5

Workout assignment engine live · Full log API (sets/reps/RPE) deployed · Workout history feed functional.

M6 **Analytics Live**
End of Week 6

Progress metrics engine deployed · Charts & trend visualisations · PDF report generation · Google Analytics integration.

M7 **Billing Live**
End of Week 7

Stripe subscription flow end-to-end · Free/Premium feature gates enforced · Email & in-app notifications · Full UI responsive QA pass.

MVP Launch
End of Week 8

All high/medium pen-test findings resolved · Performance optimisation complete · Staging → Production deploy signed off.

Critical Path

Foundation → Auth → Profiles → Workout Library → Logging → Analytics → Billing → QA → Launch
Each milestone is a hard gate. The next phase does not begin until the current milestone passes QA sign-off.

Phase Breakdown

Phase	Weeks	Key Tasks	Blocker If Late
Foundation	1	Env setup, DB schema, AWS, CI/CD	All subsequent phases
Core Features	2 & 4	Auth/RBAC, Exercise Library, Templates	Profile & Logging phases
Athlete Tools	3 & 5	Profiles, Teams, Logging, Assignment	Analytics & reporting
Analytics	6	Progress metrics, charts, PDF reports	Billing feature gates
Integrations	7	Stripe, Notifications, Feature flags	Launch readiness
QA & Launch	8	Pen test, optimisation, prod deploy	—

DELIVERABLE 04

API Endpoint Specifications

Complete REST API contracts for Athlete Profiles, Workout Logging, and Progress Tracking — ready for developer implementation.

Base URL: `https://api.evolveedge.io/v1` · **Auth:** `Authorization: Bearer <JWT>` on all endpoints · **Content-Type:** `application/json` · **Rate Limit:** 100 req/min per user · 500 req/min per org

4A — Athlete Profile Endpoints

`GET` `/athletes`

Returns paginated athlete list. Coaches see only assigned athletes. Admins see all.

Query Parameters

Param	Type	Required	Description
page	integer	No	Page number (default: 1)
limit	integer	No	Results per page (default: 20, max: 100)
search	string	No	Search by name or email
coach_id	UUID	No	Filter by assigned coach (Admin only)
status	enum	No	active | inactive | archived

RESPONSE 200 OK

```
{
  "data": [{
    "id": "uuid",
    "first_name": "Jordan",
    "last_name": "Smith",
    "email": "jordan@email.com",
    "avatar_url": "https://cdn.evolveedge.io/avatars/uuid.jpg",
    "coach_id": "uuid",
    "status": "active",
    "created_at": "2026-01-15T09:00:00Z"
  }],
  "pagination": { "page": 1, "limit": 20, "total": 84, "total_pages": 5 }
}
```

`POST` `/athletes`

Creates a new athlete. Admin or Coach only. Free tier limited to 3 athletes per coach.

REQUEST BODY

```
{
  "first_name": "Jordan",
  "last_name": "Smith",
  "email": "jordan@email.com",
  "coach_id": "uuid",
  "date_of_birth": "1998-04-22",
  "sport": "Powerlifting",
  "experience_level": "intermediate",   // beginner | intermediate | advanced | elite
  "goals": ["increase_strength", "competition_prep"]
}
```

Error Responses

Code	Reason
400	Validation failure — missing required field
409	Email already registered
403	Coach has reached Free tier roster limit (3 athletes)

GET `/athletes/{athlete_id}`

Returns full athlete profile. Coach must be assigned. Athletes can only access their own.

RESPONSE 200 OK

```
{
  "id": "uuid",
  "first_name": "Jordan", "last_name": "Smith",
  "email": "jordan@email.com",
  "sport": "Powerlifting",
  "experience_level": "intermediate",
  "coach": { "id": "uuid", "name": "Alex Turner" },
  "goals": [{
    "id": "uuid", "title": "Hit 200kg squat",
    "target_date": "2026-06-01", "status": "active"
  }],
  "biometrics": {
    "height_cm": 178, "weight_kg": 88.5,
    "body_fat_pct": 14.2, "recorded_at": "2026-03-10T08:00:00Z"
  },
  "subscription_tier": "premium",
  "status": "active",
  "last_active_at": "2026-03-18T21:45:00Z"
}
```

PATCH `/athletes/{athlete_id}`

Partial update to athlete profile. Athletes can update own fields only; Coaches update assigned athletes; Admins update all.

REQUEST BODY (ALL FIELDS OPTIONAL)

```
{
  "first_name": "Jordan",
  "phone": "+1-555-999-8888",
  "sport": "Olympic Weightlifting",
  "experience_level": "advanced",
  "biometrics": { "weight_kg": 90.1, "body_fat_pct": 13.8 }
}
```

DELETE `/athletes/{athlete_id}`

Soft-deletes (archives) an athlete. Admin only. Data retained 30 days then anonymised per HIPAA policy. Returns 409 if athlete has active assigned workloads.

POST `/athletes/{athlete_id}/goals`

Creates a goal for an athlete. Free tier limited to 1 active goal. Premium: unlimited.

REQUEST BODY

```
{
  "title": "Hit 200kg squat",
  "category": "strength",     // strength | endurance | body_comp | performance
  "target_value": 200,
  "unit": "kg",
  "target_date": "2026-06-01"
}
```

4B — Workout Logging Endpoints

POST /workout-logs

Logs a completed session. Can be self-logged or submitted by a coach on behalf of an assigned athlete (Premium only).

REQUEST BODY

```
{
  "athlete_id": "uuid",
  "workout_template_id": "uuid",
  "assignment_id": "uuid",
  "date": "2026-03-19",
  "start_time": "07:30:00",
  "end_time": "08:45:00",
  "status": "completed",        // completed | partial | skipped
  "overall_rpe": 8,             // 1-10
  "session_notes": "Felt strong on squats. Lower back tight on deadlifts.",
  "exercises": [{
    "exercise_id": "uuid",
    "exercise_name": "Back Squat",
    "order": 1,
    "sets": [
      { "set_number": 1, "reps": 5, "weight_kg": 150, "rpe": 7, "completed": true },
      { "set_number": 2, "reps": 5, "weight_kg": 155, "rpe": 8, "completed": true },
      { "set_number": 3, "reps": 4, "weight_kg": 160, "rpe": 9, "completed": true,
        "notes": "grinder" }
    ]
  },
  {
    "exercise_id": "uuid",
    "exercise_name": "Romanian Deadlift",
    "order": 2,
    "sets": [
      { "set_number": 1, "reps": 8, "weight_kg": 100, "rpe": 6, "completed": true },
      { "set_number": 2, "reps": 8, "weight_kg": 100, "rpe": 7, "completed": true },
      { "set_number": 3, "reps": 6, "weight_kg": 100, "rpe": 8, "completed": false,
        "skip_reason": "lower_back_fatigue" }
    ]
  }],
  "media_urls": ["https://cdn.evolveedge.io/logs/video_uuid.mp4"]  // Premium only
}
```

RESPONSE 201 CREATED

```
{
  "id": "uuid",
  "athlete_id": "uuid",
  "date": "2026-03-19",
  "duration_minutes": 75,
  "total_volume_kg": 4180,
  "exercises_completed": 2,
  "sets_completed": 5,
  "sets_skipped": 1,
  "overall_rpe": 8,
  "created_at": "2026-03-19T09:00:00Z"
}
```

Field Definitions

Field	Type	Notes
overall_rpe	integer 1–10	Rate of Perceived Exertion for session
weight_kg	decimal	Must be > 0; use 0 for bodyweight exercises
skip_reason	enum	time | fatigue | injury | lower_back_fatigue | other
media_urls	array	Premium only · max 5 files · 100MB each

GET `/workout-logs`

Returns paginated workout history. Scoped by role. Free tier: from_date cannot exceed 30 days ago — returns 403 "upgrade_required" if exceeded.

Param	Type	Description
athlete_id	UUID	Filter by athlete (Coach/Admin only)
from_date	date	ISO 8601 range start
to_date	date	ISO 8601 range end
exercise_id	UUID	Filter logs containing specific exercise
status	enum	completed | partial | skipped
page / limit	integer	Pagination — max 50 per page

`PATCH` `/workout-logs/{log_id}`

Updates a logged session. Athletes (Free) can edit within 24 hours only. Coaches and Admins have no time restriction. Returns 403 if Free tier athlete exceeds 24-hour window.

`POST` `/workout-logs/{log_id}/media`

Attaches media to a log entry. PREMIUM ONLY

Field	Type	Notes
file	binary	multipart/form-data · MP4, MOV, JPG, PNG
type	enum	video | image
label	string	e.g., "Squat set 3 side view"

Limits: max 5 files, 100MB per file, 500MB per log entry total.

4C — Progress Tracking Endpoints

GET `/athletes/{athlete_id}/progress/summary`

High-level progress summary card for dashboard display. Includes streaks, PRs, and goal progress.

RESPONSE 200 OK

```
{
  "athlete_id": "uuid",
  "period": "last_30_days",
  "total_sessions": 18,
  "total_volume_kg": 74200,
  "avg_session_duration_minutes": 68,
  "avg_rpe": 7.4,
  "workouts_completed_pct": 90,
  "personal_records": [{
    "exercise_id": "uuid",
    "exercise_name": "Back Squat",
    "value_kg": 182.5,
    "achieved_date": "2026-03-15"
  }],
  "streak": { "current_days": 12, "longest_days": 21 },
  "goal_progress": [{
    "goal_id": "uuid",
    "title": "Hit 200kg squat",
    "current_value": 182.5,
    "target_value": 200,
    "progress_pct": 91.25
  }]
}
```

GET /athletes/{athlete_id}/progress/strength

Strength progression data over time for charting. PREMIUM ONLY

Param	Type	Description
exercise_id	UUID	Required — filter by exercise
metric	enum	1rm_estimated | max_weight | total_volume
from_date / to_date	date	Range for data points
interval	enum	day | week | month

RESPONSE 200 OK

```
{
  "exercise_name": "Back Squat",
  "metric": "1rm_estimated",
  "unit": "kg",
  "data_points": [
    { "date": "2026-01-06", "value": 170.0 },
    { "date": "2026-01-20", "value": 175.0 },
    { "date": "2026-03-17", "value": 182.5 }
  ],
  "trend": "increasing",
  "change_pct": 7.35,
  "pr": { "value": 182.5, "date": "2026-03-17" }
}
```

GET /athletes/{athlete_id}/progress/volume

Weekly or monthly training volume (sets × reps × weight) for load management and overtraining prevention.

RESPONSE 200 OK

```
{
  "interval": "week",
  "unit": "kg",
  "data_points": [
    { "week_start": "2026-01-06", "total_volume_kg": 18400, "sessions": 4 },
    { "week_start": "2026-01-13", "total_volume_kg": 21200, "sessions": 5 },
    { "week_start": "2026-01-20", "total_volume_kg": 19800, "sessions": 4 }
  ],
  "avg_weekly_volume_kg": 19800,
  "trend": "stable"
}
```

GET /athletes/{athlete_id}/progress/body-composition

Body composition trend (weight, body fat %, lean mass) over time. PREMIUM ONLY

RESPONSE 200 OK

```
{
  "data_points": [
    { "recorded_at": "2026-01-01", "weight_kg": 90.2, "body_fat_pct": 15.8, "lean_mass_kg":
75.9 },
    { "recorded_at": "2026-02-01", "weight_kg": 89.1, "body_fat_pct": 14.9, "lean_mass_kg":
75.8 },
    { "recorded_at": "2026-03-01", "weight_kg": 88.5, "body_fat_pct": 14.2, "lean_mass_kg":
75.9 }
  ],
  "trend": {
    "weight_change_kg": -1.7,
    "body_fat_change_pct": -1.6,
    "lean_mass_change_kg": 0.0
  }
}
```

GET `/athletes/{athlete_id}/progress/report`

Generates structured progress report. Returns JSON or downloadable PDF. PREMIUM ONLY

Param	Type	Description
from_date / to_date	date	Report period
format	enum	json | pdf (default: json)
include_sections	array	summary, strength, volume, body_comp, goals, recommendations

PDF response: `Content-Type: application/pdf` with `Content-Disposition: attachment`.

POST `/athletes/{athlete_id}/progress/biometrics`

Records a new biometric measurement. Returns entry with calculated lean_mass_kg.

REQUEST BODY

```
{
  "weight_kg": 88.5,
  "body_fat_pct": 14.2,
  "height_cm": 178,
  "recorded_at": "2026-03-19T08:00:00Z",
  "notes": "Morning, fasted measurement"
}
```

GET /progress/leaderboard

Relative performance comparison across a coach's roster. PREMIUM ONLY

Param	Description
coach_id (UUID)	Required — returns that coach's athletes
metric	volume | sessions | streak
period	week | month

RESPONSE 200 OK

```
{
  "metric": "total_volume_kg",
  "period": "month",
  "rankings": [
    { "rank": 1, "athlete_id": "uuid", "name": "Jordan S.", "value": 74200 },
    { "rank": 2, "athlete_id": "uuid", "name": "Sam K.",    "value": 68400 },
    { "rank": 3, "athlete_id": "uuid", "name": "Riley M.",  "value": 61000 }
  ]
}
```

API Error Reference

HTTP Code	Error Key	Meaning
400	validation_error	Missing or invalid request field
401	unauthorized	Missing or expired JWT token
403	forbidden	Valid token but insufficient role or tier
403	upgrade_required	Feature is gated behind Premium subscription
404	not_found	Resource does not exist or not accessible
409	conflict	Resource conflict (e.g. duplicate email, active workloads)
429	rate_limit_exceeded	Too many requests — retry after cooldown
500	internal_error	Server-side failure — log and retry with exponential backoff

Document Complete. All four deliverables are ready for insertion into the Evolve Edge MVP Requirements Document and direct handoff to your development team.